

LeetCode 1614: Maximum Nesting Depth of the Parentheses

Visual Logic & Approach Breakdown

Objective: Calculate the maximum nesting depth of valid parentheses within a string.

Step-by-Step Execution:

- Initialize `ans = 0` to record the maximum depth reached.
- Initialize `currsz = 0` and a character stack `st` to track current nesting level.
- Iterate through every character `c` in the string:
 - If `c == '('` : Push to the stack, increase `currsz`, and update `ans = max(ans, currsz)`.
 - If `c == ')'` : Pop from the stack (if it contains elements) and decrease `currsz`.
 - Ignore any other alphanumeric characters or symbols.

Source Code (C++)

```
class Solution {
public:
    int maxDepth(string s) {

        int ans=0;
        int currsz=0;
        stack<char> st;

        for(char c:s){
            if(c=='('){
                st.push(c);
                currsz++;
                ans=max(ans,currsz);
            }
            else if(c==')'){
                if(!st.empty()){
                    st.pop();
                    currsz--;
                }
            }
            else continue;
        }

        return ans;
    }
};
```